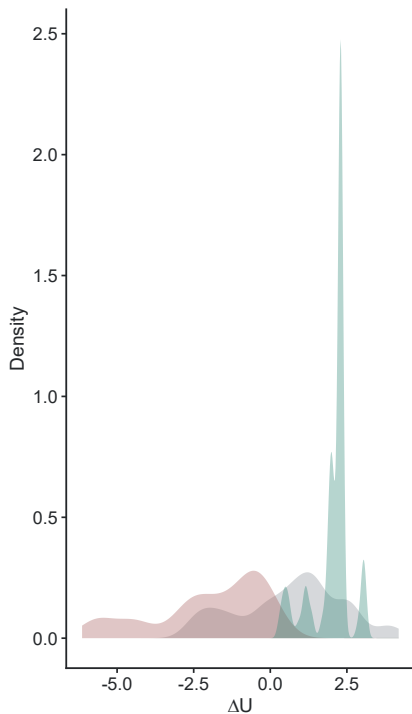


Supplementary Figure S6 | ΔU construction and orientation audit

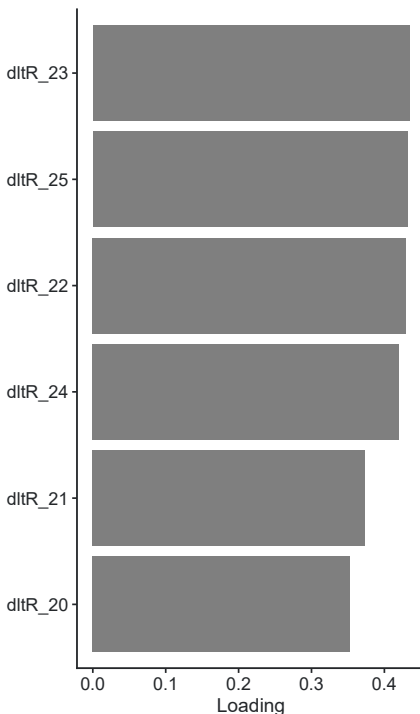
Sign is oriented by the controlled leading-versus-competing contrast; it is not an independent target-free label

a Oriented PC1 score by phase

Phase ■ critical ■ negative ■ positive



b PC1 loading orientation



c Alternative ΔU scaling audit

Band label ● critical ● negative ● positive

